

SentinelOps AI

Multi-Agent Cloud Incident Response Platform

Junior FDE Pre-screening Assignment | Multi-Agent System Design & Implementation

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Live: sentinelops-ai-427637329209.us-central1.run.app

1. Multi-Agent Architecture

SentinelOps AI is a cloud-hosted Security Operations Center (SOC) simulation platform designed to demonstrate how multiple AI agents can collaborate safely during cloud incident investigations. The platform follows a shared-context orchestration model where each agent performs a specialized task and passes enriched workflow state to the next agent through a centralized orchestration layer.

Agent Pipeline

#	Agent	Responsibility
1	Intake	Normalizes unstructured incident text into intent, entities, affected services, and severity classification
2	Planner	Emits ordered task list and determines execution phases for downstream agents
3	Log Analysis	Joins cloud_logs.csv and security_events.csv to surface evidence from telemetry
4	Compliance	Enforces compliance_rules.json and injection heuristics; can short-circuit pipeline before remediation
5	Root Cause	Applies rule-based and LLM inference over evidence bundles to identify probable cause
6	Remediation	Produces gated action list with rationale; skipped entirely on compliance block
7	Validation	Post-remediation scan for high-risk language (delete, purge, bypass, PII exposure); sets requires_approval
8	Auditor	Aggregates confidence scores, writes executive summary, and finalizes immutable audit trail

Agents execute sequentially for explainability and auditability. The Compliance and Validation Agents can stop workflows early if unsafe actions are detected.

2. Security, Safety & Guardrails

SentinelOps AI uses deterministic guardrails outside the LLM to reduce unsafe AI behavior. The system includes multiple layers of protection:

- Role-based access control for SOC Analyst, SOC Manager, Incident Commander, Compliance Reviewer, and Observer roles
- Temporary access elevation requests requiring SOC Manager approval
- Human-in-the-loop approvals for sensitive remediation actions
- Input validation to block unsafe prompts such as deleting production databases or bypassing compliance rules
- Output filtering to redact PII-like patterns and dangerous remediation instructions
- Full audit logging for logins, approvals, blocked actions, and investigations

All incident data used in the project is synthetic. No real production telemetry or personally identifiable information (PII) is processed. API keys are securely stored as environment variables in Google Cloud Run rather than hardcoded in the source code.

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3. Implementation Approach

Tech Stack

Component	Technology
Language	Python 3.11
UI Framework	Streamlit (dark enterprise theme)
Primary LLM	Groq API — llama-3.3-70b-versatile
Fallback LLM	OpenAI API — GPT-4o-mini
Final Fallback	Deterministic mock (rules-based, keyword matching)
Cloud Deployment	Google Cloud Run — public HTTPS endpoint
Alt Deployment	Azure App Service — documented in <code>azure_deployment_guide.md</code>
Secret Management	Environment variables (Cloud Run / App Service Application Settings)
Version Control	GitHub — public repository
Data Layer	In-memory: <code>cloud_logs.csv</code> , <code>security_events.csv</code> , <code>compliance_rules.json</code>

A custom orchestration layer was intentionally used instead of frameworks such as LangChain because the workflow is deterministic, audit-sensitive, and easier to explain using lightweight orchestration.

The system also includes retry handling and fallback logic if AI services become unavailable.

4. Use of AI/LLMs & Agent Collaboration

Large Language Models (LLMs) are used for:

- Incident summarization
- Task planning
- Root cause reasoning
- Evidence analysis
- Remediation recommendation generation
- Executive report creation

Safety enforcement is intentionally separated from the LLM. Compliance validation, access control, and dangerous action detection are handled deterministically outside the AI model.

Agents collaborate through a shared incident context object. The Compliance Agent can veto unsafe workflows before remediation begins, while the Validation Agent acts as a critic layer that reviews remediation recommendations before final reporting.

The system deliberately balances AI autonomy with human oversight. Sensitive operations such as production rollback, firewall modification, or data purge actions always require SOC Manager approval.